

Topics: Review of derivatives and integrals.

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1. Find the derivative of $f(x) = \cos(x^2 + e^x)$.

By the Chain Rule,

$$f'(x) = -\sin(x^2 + e^x)(2x + e^x).$$

2. Given that $x \ln(y) + e^y = 3$, find $\frac{dy}{dx}$ by implicit differentiation.

Differentiate both sides with respect to x :

$$\frac{d}{dx}(x \ln(y)) + \frac{d}{dx}(e^y) = 0.$$

Using the Product Rule and Chain Rule,

$$\ln(y) + x \cdot \frac{1}{y} \frac{dy}{dx} + e^y \frac{dy}{dx} = 0.$$

Collect the $\frac{dy}{dx}$ terms:

$$\left(\frac{x}{y} + e^y\right) \frac{dy}{dx} = -\ln(y).$$

Therefore,

$$\frac{dy}{dx} = \frac{-\ln(y)}{\frac{x}{y} + e^y} = \frac{-y \ln(y)}{x + ye^y}.$$

3. Suppose that $f(x)$ is a differentiable function such that $f(1) = 7$ and $f'(1) = 4$. Find an equation of the tangent line to $y = \sqrt{4 + 3f(x)}$ at $x = 1$.

Let $h(x) = \sqrt{4 + 3f(x)} = (4 + 3f(x))^{1/2}$.

By the Chain Rule, we get

$$h'(x) = \frac{1}{2}(4 + 3f(x))^{-1/2} \cdot 3f'(x) = \frac{3f'(x)}{2\sqrt{4 + 3f(x)}}.$$

Evaluate at $x = 1$:

$$h(1) = \sqrt{4 + 3f(1)} = \sqrt{4 + 3 \cdot 7} = \sqrt{25} = 5, \quad h'(1) = \frac{3f'(1)}{2\sqrt{4 + 3f(1)}} = \frac{3 \cdot 4}{2 \cdot 5} = \frac{6}{5}.$$

Therefore the tangent line at $x = 1$ is

$$y - 5 = \frac{6}{5}(x - 1) \quad \text{OR} \quad y = \frac{6}{5}x + \frac{19}{5}.$$

4. Find all critical points of

$$f(x) = x^3 - 3x^2 - 9x + 2$$

and determine whether each is a local maximum or a local minimum.

First, we find the derivative:

$$f'(x) = 3x^2 - 6x - 9 = 3(x+1)(x-3).$$

Thus, the critical numbers are $x = -1$ and $x = 3$.

To classify the critical points, we use the Second Derivative Test. We have

$$f''(x) = 6x - 6.$$

At $x = -1$,

$$f''(-1) = -12 < 0,$$

so f has a local maximum at $x = -1$. Since

$$f(-1) = (-1)^3 - 3(-1)^2 - 9(-1) + 2 = 7,$$

the local maximum occurs at $(-1, 7)$.

At $x = 3$,

$$f''(3) = 12 > 0,$$

so f has a local minimum at $x = 3$. Since

$$f(3) = 3^3 - 3(3)^2 - 9(3) + 2 = -25,$$

the local minimum occurs at $(3, -25)$.

5. Evaluate the indefinite integrals.

(a) $\int e^{\tan x} \sec^2(x) dx$

We use u -substitution. Let $u = \tan x$, so $du = \sec^2(x)dx$. Now,

$$\int e^{\tan x} \sec^2(x) dx = \int e^u du = e^u + C = e^{\tan x} + C$$

(b) $\int (2x^4 + 1) \ln(x) dx$

We use integration by parts:

$$\begin{array}{ll} u = \ln(x) & du = \frac{1}{x} dx \\ dv = (2x^4 + 1) dx & v = \frac{2x^5}{5} + x \end{array}$$

Now, applying integration by parts:

$$\begin{aligned}\int (2x^4 + 1) \ln(x) dx &= \left(\frac{2x^5}{5} + x\right) \ln(x) - \int \left(\frac{2x^5}{5} + x\right) \cdot \frac{1}{x} dx \\ &= \left(\frac{2x^5}{5} + x\right) \ln(x) - \int \left(\frac{2x^4}{5} + 1\right) dx \\ &= \left(\frac{2x^5}{5} + x\right) \ln(x) - \left(\frac{2x^5}{25} + x\right) + C\end{aligned}$$

6. Evaluate the definite integrals.

(a) $\int_0^{\pi/2} \sin^2 \theta \cos \theta d\theta$

We use u -substitution. Let $u = \sin \theta$, so $du = \cos \theta d\theta$.

Change the limits: when $\theta = 0$, $u = \sin 0 = 0$; when $\theta = \pi/2$, $u = \sin(\pi/2) = 1$.

$$\int_0^{\pi/2} \sin^2 \theta \cos \theta d\theta = \int_0^1 u^2 du = \frac{1}{3} u^3 \Big|_0^1 = \frac{1}{3}.$$

(b) $\int_0^3 \frac{x}{\sqrt{x^2 + 1}} dx$

We use u -substitution. Let $u = x^2 + 1$, so $du = 2x dx$ and $\frac{1}{2} du = x dx$.

Change the limits: when $x = 0$, $u = 1$; when $x = 3$, $u = 10$.

$$\int_0^3 \frac{x}{\sqrt{x^2 + 1}} dx = \frac{1}{2} \int_1^{10} u^{-1/2} du = \frac{1}{2} 2u^{1/2} \Big|_1^{10} = \sqrt{u} \Big|_1^{10} = \sqrt{10} - 1.$$

7. A particle is moving with acceleration at time t given by

$$a(t) = 3 \cos t - 2 \sin t.$$

Given that $s(0) = 0$ and $v(0) = 4$, determine the position of the particle $s(t)$ at any time t .

$$v(t) = \int a(t) dt = \int 3 \cos t - 2 \sin t dt = 3 \sin t + 2 \cos t + C$$

$$v(0) = 2 \cos(0) + C = 4 \Rightarrow C = 2$$

$$v(t) = 3 \sin t + 2 \cos t + 2$$

$$s(t) = \int v(t) dt = \int 3 \sin t + 2 \cos t + 2 dt = -3 \cos t + 2 \sin t + 2t + C'$$

$$s(0) = -3 \cos(0) + C' = 0 \Rightarrow C' = 3$$

$$s(t) = -3 \cos(t) + 2 \sin(t) + 2t + 3$$

8. Consider the region R bounded by the curves $x = y^2$ and $y = \frac{1}{2}x$.

(a) Sketch the region R .

First, we find the points where the curves intersect. Since $y = \frac{1}{2}x$, we have $x = 2y$.

Substituting $x = 2y$ into $x = y^2$ gives

$$2y = y^2.$$

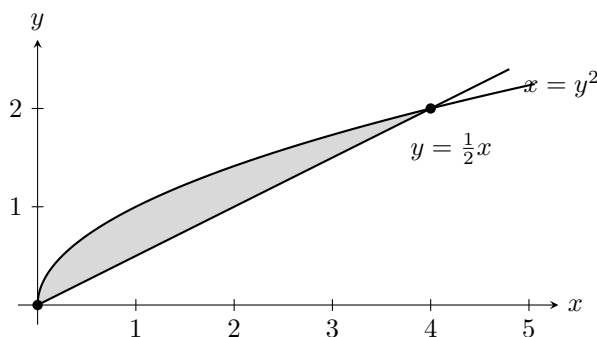
Rearranging and factoring, we get

$$y(y - 2) = 0.$$

Thus, $y = 0$ or $y = 2$, giving the intersection points

$$(0, 0) \quad \text{and} \quad (4, 2).$$

The curve $x = y^2$ is a parabola opening to the right and $y = \frac{1}{2}x$ is a line through the origin with slope $\frac{1}{2}$. Thus the region R is as follows.



(b) Find the area of the region R .

It is easiest to integrate with respect to y . For $0 \leq y \leq 2$, the left boundary is $x = y^2$ and the right boundary is $x = 2y$. Therefore,

$$\begin{aligned} A &= \int_0^2 (2y - y^2) dy \\ &= \left(y^2 - \frac{y^3}{3} \right) \Big|_0^2 \\ &= 4 - \frac{8}{3} \\ &= \frac{4}{3}. \end{aligned}$$

Thus, the area of R is $\frac{4}{3}$.

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Differentiation Formulas

$$\frac{d}{dx}(c) = 0$$

$$\frac{d}{dx}(b^x) = b^x \ln b$$

$$\frac{d}{dx}(f(x) + g(x)) = f'(x) + g'(x)$$

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(cf(x)) = cf'(x)$$

$$\frac{d}{dx}(f(x) - g(x)) = f'(x) - g'(x)$$

$$\frac{d}{dx}(e^x) = e^x$$

The Product Rule

$$\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$$

The Quotient Rule

$$\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$$

The Chain Rule

$$\frac{d}{dx}(f(g(x))) = f'(g(x))g'(x)$$

Derivatives of Trigonometric Functions

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

The Chain Rule for the power of a function

$$\frac{d}{dx}[f(x)^n] = n[f(x)]^{n-1} f'(x)$$

Derivatives of Inverse Trigonometric Functions

$$\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx}(\arccos x) = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\operatorname{arccot} x) = -\frac{1}{1+x^2}$$

Derivatives of Logarithmic Functions

$$\frac{d}{dx}(\log_b x) = \frac{1}{x \ln b}$$

$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

$$\frac{d}{dx}[\ln f(x)] = \frac{f'(x)}{f(x)}$$

Integration Formulas

a. If $n \neq -1$, $\int x^n dx = \frac{x^{n+1}}{n+1} + C$

b. $\int e^x dx = e^x + C$

c. $\int \sin x dx = -\cos x + C$

d. $\int \sec^2 x dx = \tan x + C$

e. $\int \sec x \tan x dx = \sec x + C$

f. $\int \sec x dx = \ln |\sec x + \tan x| + C$

g. $\int \tan x dx = \ln |\sec x| + C$

h. $\int \frac{1}{x} dx = \ln |x| + C$

i. $\int b^x dx = \frac{b^x}{\ln b} + C$

j. $\int \cos x dx = \sin x + C$

k. $\int \csc^2 x dx = -\cot x + C$

l. $\int \csc x \cot x dx = -\csc x + C$

m. $\int \csc x dx = -\ln |\csc x + \cot x| + C$

n. $\int \cot x dx = \ln |\sin x| + C$

where C is any real constant.

The Substitution Rule: This rule is useful when the integrand contains a function and (a constant multiple of) its derivative. Let $u = g(x)$, so $du = g'(x) dx$. Then

$$\int f(g(x)) g'(x) dx = \int f(u) du.$$

Integration by Parts: This rule is useful for integrals that look like a product of two functions. Choose u and dv from the integrand so that $du = u' dx$ and $v = \int dv$. Then

$$\int u dv = uv - \int v du.$$